

Tanvir Ahmed Shanto

Ph.D. in Mechanical Engineering

Experimental & Computational Solid Mechanics | Design of Experiment | Material Characterization | Process Innovation

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Research Profile

Mechanical Engineering Ph.D. with more than seven years of combined research and engineering experience in experimental and computational solid mechanics, fracture mechanics, statistical inference, and data-driven manufacturing. Doctoral research integrates factorial experimental design, infrared thermography, X-ray micro-computed tomography, quasi-static and viscoelastic testing, finite element analysis, and machine learning to quantify and validate process–structure–property relationships in additively manufactured polymers and polymer composites.

Education

Ph.D. in Mechanical Engineering

Sep 2021–Aug 2026

The University of Texas at Arlington

Arlington, Texas, USA

Dissertation: Optimizing Interlayer Adhesion in Fused Filament Fabrication (FFF) for Improved Mechanical Performance

Advisor: Dr. Robert M. Taylor

B.Sc. in Naval Architecture and Marine Engineering

Feb 2013–Sep 2017

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

Thesis: Predicting crack propagation at vulnerable locations of ship structures and analyzing possible mitigation measures

Research Experience

Graduate Student Researcher

Aug 2021–Aug 2026

The University of Texas at Arlington

Arlington, Texas, USA

- Developed an experimentally anchored process–structure–property framework for thermally assisted FFF by integrating in-process MWIR thermography, X-ray μ -CT, tensile testing, dynamic mechanical analysis, Mode-I compact-tension testing, and SEM fractography to connect interfacial thermal history and void morphology with structural response.
- Designed and analyzed a randomized $2 \times 3 \times 5$ full-factorial experiment (60 print batches) using GLM/ANOVA, Type III sums of squares, interaction analysis, residual normality/variance/independence diagnostics, and Tukey HSD; identified a printhead–speed interaction and a sample-spacing effect on toughness.
- Demonstrated that controlled in-situ annealing increased ABS-CF ultimate tensile strength by 39%, elastic modulus by 30%, glassy storage modulus by 63%, and Mode-I interlaminar fracture resistance by approximately 92%. Representative μ -CT measurements showed crack-path void content decreasing from approximately 4.0% to 0.7%, accompanied by a shift from interfacial delamination toward cohesive fracture.
- Developed μ -informed RVE and cohesive-zone models in Abaqus for DCB and SENB specimens to estimate direction-dependent elastic properties, characterize interfacial traction–separation behavior, and compare simulated load–displacement response with experiments.
- Designed and prototyped adaptive in-situ annealing printhead hardware using GD&T and DFM/DFA principles, contributing to one issued and one pending U.S. patent and translating the work into peer-reviewed publications and commercialization activities.
- Executed ASTM-aligned tensile, compression, three-point-bending, fatigue, DMA, and fracture-testing campaigns; mentored students in test planning, instrumentation, data reduction, statistical interpretation, and technical writing.

Research and Technical Expertise

Computational mechanics	Abaqus, ANSYS, HyperWorks, HyperStudy, and CAESAR II; finite element analysis, multiphysics analysis, experimental-to-model comparison, RVE/micromechanics, and cohesive-zone methods.
Experimental mechanics	Tensile, compression, three-point-bending, fatigue, DMA, and Mode-I fracture testing.
NDE & characterization	MWIR/IR thermography; X-ray μ -CT; optical and SEM/EDS fractography; DSC, TGA, STA, FTIR, XRD, and AFM.
Statistics & UQ foundation	Full-factorial and fractional-factorial DoE; GLM/ANOVA.
Data science & ML	Python, MATLAB, SAS Studio, Fortran, and LLM few-shot benchmarking.
Design & manufacturing	SolidWorks, AutoCAD/Plant 3D, ASME Y14.5 GD&T, DFM/DFA; FFF, SLA, SLS, material jetting, DLP, multi-axis extrusion, DFAM, and LPBF; ASTM and ISO test methods.

Peer-Reviewed Journal Articles

1. **Shanto, T. A.**, Ahmed, R., Patel, P., Zulqernine, M. J., Chen, V., Jain, A., & Taylor, R. M. (2026). In-situ annealing and thermal-morphological evolution during extrusion-based additive manufacturing. *Journal of Manufacturing Processes*, 170, 303–318. doi:10.1016/j.jmapro.2026.05.005
2. Ahmed, R., Dola, I. S., Ahmed, S., Rahman, M. M., **Shanto, T. A.**, & Taylor, R. M. (2026). A review of design and fabrication strategies for thermoplastic lightweight thin-walled stiffened shells and lattice structures utilizing fused-filament fabrication (FFF). *International Journal of Advanced Manufacturing Technology*. doi:10.1007/s00170-026-18078-6
3. Patel, P., Ahmed, R., **Shanto, T. A.**, Jain, A., & Taylor, R. M. (2025). Experimental characterization of enhanced fused filament fabrication (FFF) of tall thin-walled structures using polylactic acid (PLA). *International Journal of Advanced Manufacturing Technology*, 139(11–12), 5663–5675. doi:10.1007/s00170-025-16171-w
4. Uddin, M. M., Akhtaruzzaman, R., **Shanto, T. A.**, & Hasan, M. N. (2023). Study the performance of a dynamic wall heat exchanger using computational fluid dynamics. *European Journal of Engineering and Technology Research*, 8(3), 38–42. doi:10.24018/ejeng.2023.8.3.3043
5. Patel, P., Rahman, M. M., **Shanto, T. A.**, Kulkarni, P., Ahmed, R., Jain, A., & Taylor, R. M. (2026). Evaluation of in process heating for extrusion printing of tall thin-walled structures. *Progress in Additive Manufacturing*. doi:10.1007/s40964-026-01852-z

Manuscripts Under Review

1. **Shanto, T. A.**, Rahman, M. M., Barua, S., Ahmed, S., Raiyan, A. Z., Ahmed, R., Jain, A., Adnan, A., & Taylor, R. M. In-situ annealing of vertically printed short-carbon-fiber ABS: A process–structure–property study of bonding potential, voids, and z-direction performance. Under Review at *Composites Part B: Engineering*.
2. **Shanto, T. A.**, Ahmed, R., Patel, P., Barua, S., Dola, I. S., Chen, V., Jain, A., & Taylor, R. M. Adaptive annealing in extrusion-based additive manufacturing and machine-learning-based prediction of mechanical performance. Under Review at *Additive Manufacturing*.
3. Ahmed, R., **Shanto, T. A.**, Ahmed, S., Raiyan, A. Z., Adnan, A., Jain, A., & Taylor, R. M. Process–structure–property relationships in FFF PLA parts. Under Review at *Progress in Additive Manufacturing*.

Works in Progress

1. **Shanto, T. A.**, Ahmed, S., Labanya, F. N., Lin, S., & Taylor, R. M. Characterizing and improving fracture behavior in 3D-printed polymers using Mode-I fracture testing. Manuscript under internal review.
2. Rahman, M. M., **Shanto, T. A.**, Ahmed, R., & Taylor, R. M. Improving the sealing capacity of FFF ducts for aerospace environmental-control-system applications. Manuscript in preparation.
3. Ahmed, S., **Shanto, T. A.**, Ahmed, R., & Taylor, R. M. Improving the mechanical performance of lightweight PLA for unmanned aerial vehicles. Ongoing research.
4. Ahmed, R., Zulqernine, M. J., **Shanto, T. A.**, Lin, S., & Taylor, R. M. Improving the compressive strength of thin-walled stiffened cylindrical structures and characterizing damage using acoustic-emission sensing. Ongoing research.

Peer-Reviewed Conference Publications

1. Dola, I. S., Ahmed, R., Zulqernine, M. J., **Shanto, T. A.**, & Taylor, R. M. (2026). Evaluation of deep learning architectures for defect detection in fused filament fabrication. *SAMPE Conference and Exhibition*. doi:10.33599/nasampe/s.26.91
2. Ahmed, S., Rahman, M. M., **Shanto, T. A.**, Ahmed, R., & Taylor, R. M. (2026). A statistical approach for evaluating printing-temperature effects on lightweight and generic polylactic acid in fused filament fabrication. *IISE Annual Conference Proceedings*. [Link](#)

3. Rahman, M. M., Ahmed, S., **Shanto, T. A.**, Ahmed, R., & Taylor, R. M. (2026). A statistical study on the influence of build-volume temperature and bead overlapping of PEI in fused filament fabrication. *IISE Annual Conference Proceedings*. [Link](#)
4. Ahmed, R., **Shanto, T. A.**, Rahman, M. M., Taylor, R. M., & Jain, A. (2025). Machine learning-based prediction of toughness in fused filament fabrication: Leveraging in-process annealing with enhanced printheads. In *Solid Freeform Fabrication 2025: Proceedings of the 36th Annual International Solid Freeform Fabrication Symposium – An Additive Manufacturing Conference*, 1182–1200. [PDF](#)
5. **Shanto, T. A.**, Pavel, H. R., Ahmed, R., Abdullah, M., & Taylor, R. M. (2025). Leveraging large language models for process parameter optimization in 3D-printed ABS polymer specimens. *IISE Annual Conference and Expo 2025, Conference Proceedings*, 1351–1356. [doi:10.21872/2025IISE_6901](#)
6. **Shanto, T. A.**, Shahriar, M. A., Ahmed, T., Zulqernine, M. J., & Taylor, R. M. (2025). Predicting mechanical strength in FDM-printed ABS parts with in-process annealing: A machine-learning approach. *IISE Annual Conference and Expo 2025, Conference Proceedings*, 1224–1229. [doi:10.21872/2025IISE_6734](#)
7. Abdullah, M., Ozay, D., Ahmed, S. M. T., **Shanto, T. A.**, & Sridhar, E. P. (2025). Genetic algorithms in order-picking route optimization: A review of advances and implications for logistics. *IISE Annual Conference and Expo 2025, Conference Proceedings*, 1375–1380. [doi:10.21872/2025IISE_6932](#)
8. Ahmed, R., Niloy, R. S., Mozumder, M. R., & **Shanto, T. A.** (2024). Application of fused filament fabrication in the marine sector: From rapid prototyping to final product. *Proceedings of MARTEC 2024*. [PDF](#)
9. Zulqernine, M. J., Alam, M. A., Uddin, M. R., Dola, I. S., & **Shanto, T. A.** (2024). An investigation on the applications of additive manufacturing in the marine industry. *Proceedings of MARTEC 2024*. [PDF](#)

Patents

1. Taylor, R. M., Rane, R., Mrinal, M., Patel, P. T., & **Shanto, T. A.** *Fused filament fabrication print-head system*. U.S. Patent 12,496,776 B2. [Patent record](#)
2. **Shanto, T. A.**, Taylor, R. M., Mrinal, M., Ahmed, R., Patel, P. T., Rahman, M. M., Dola, I. S., & Deshpande, R. *Fused filament fabrication adaptive annealing print-head system*. U.S. Patent Application No. 19/660,770.

Teaching and Mentoring Experience

Teaching Assistant

The University of Texas at Arlington

Sep 2021–Present

Arlington, Texas, USA

- **Additive Manufacturing (MAE 5329; 200+ students):** Conducted laboratory exercises in SLA, SLS, material jetting, and FFF; trained students on universal testing, Simplify3D, PolyWorks, 3-Matic, and FARO arm scanning; mentored senior-design teams through 3D-printed UAV design, fabrication, testing, and competition activities.
- **Solid Mechanics (MAE 2312; 30 students):** Led supplemental problem-solving sessions, supported coursework and projects, held office hours, and evaluated assignments, quizzes, and examinations.
- **Additional courses:** Problems in MAE (MAE 1140; 100 students), Introduction to Manufacturing (MAE 3344; 50 students), Thermodynamics I (MAE 3310; 40 students), and Engineering Analysis (MAE 5332; 48 students); supported instruction, assessment, office hours, and project guidance.
- Selected for the I-ENGAGE Student Mentorship Program to guide undergraduate researchers in technical concepts, research methods, and problem-solving.

Professional and Industrial Experience

Co-Founder and Technical Lead

Increscent, Inc.

Aug 2022–Present

Texas, USA

- Translated a university-developed extrusion-based 3D-printing method into a staged R&D and commercialization roadmap with measurable performance milestones and qualification criteria.
- Authored technical proposals and grant applications, including a successful \$50,000 NSF I-Corps Team award and two SBIR Phase I proposals on AI-enabled extrusion-based additive manufacturing.
- Led structured customer discovery and translated voice-of-customer evidence into product requirements; prepared technical white papers, test reports, pitch decks, and partner/investor materials.

Mechanical Engineer

Oct 2018–Oct 2019

Super Petrochemical Limited

Bangladesh

- Designed process equipment and piping layouts in SolidWorks and AutoCAD Plant 3D, performed piping-stress analysis in CAESAR II, and supported proposal review for a \$10M mooring and pipeline program.
- Coordinated a 70-person mechanical workshop and maintenance crew, prioritized work orders, allocated resources, and enforced safe, high-quality repairs using Lean Six Sigma practices.
- Reduced oil-tanker construction and maintenance costs by approximately 10% through process-waste identification, planning improvements, and vendor coordination.

Honors, Grants, and Awards

- **NSF I-Corps Team Award, \$50,000:** High-Performance 3D-Printed Plastic Parts Through Novel Process Enhancement.
- **Maverick Business Pitch Competition, \$35,000 venture award:** Recognized for technical innovation, commercialization strategy, and business presentation.
- **Wheelwright Foundation Scholarship, \$2,000 (2023).**
- **I-ENGAGE Student Mentorship Program, \$2,000 stipend:** Selected to mentor undergraduate researchers.
- **Blackstone LaunchPad Campus Award, \$1,000 (2022).**

Professional Service and Membership

Reviewer, *Journal of Thermal Analysis and Calorimetry* (Springer Nature), 2026 — 1 manuscript.

Member, American Society of Mechanical Engineers (ASME).

Academic References

Dr. Robert M. Taylor

Professor in Practice

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